Name: Quiz 2 — 7/9

1. Compute $\dfrac{d}{dx}\arctan(x^3+1/x)$.

Solution. We use the chain rule:

$$\frac{d}{dx}\arctan(x^3+1/x) = \frac{1}{1+(x^3+1/x)^2}\cdot(3x^2-1/x^2).$$

2. Compute $\lim_{x\to 1}\dfrac{x^7-1}{x^4-1}$.

Solution. Since plugging in $x=1$ gives a zero in the numerator and denominator, we're looking at an indeterminate form of type $0/0$ and can use l'Hôpital's rule:

$$\lim_{x\to 1}\frac{x^7-1}{x^4-1} = \lim_{x\to 1}\frac{7x^6}{4x^3} = \frac{7}{4}.$$

3. Compute $\int x^2\sin(2x)\,dx$.

Solution. We integrate by parts, taking $u=x^2$ and $dv=\sin(2x)$ to get $du=2x$ and $v=-(1/2)\cos(2x)$. Then

$$\begin{aligned}\int x^2\sin(2x)\,dx &= -\frac{1}{2}x^2\cos(2x) - \int 2x\cdot(-\frac{1}{2}\cos(2x))\,dx\\ &= -\frac{1}{2}x^2\cos(2x) + \int x\cos(2x)\,dx,\end{aligned}$$

and we can integrate by parts a second time. This time, $u=x$ and $dv=\cos(2x)$ gives us $du=1$ and $v=(1/2)\sin(2x)$ for

$$\begin{aligned}\int x^2\sin(2x)\,dx &= -\frac{1}{2}x^2\cos(2x) + \int x\cos(2x)\,dx\\ &= -\frac{1}{2}x^2\cos(2x) + \frac{1}{2}x\sin(2x) - \int \frac{1}{2}\sin(2x)\,dx\\ &= -\frac{1}{2}x^2\cos(2x) + \frac{1}{2}x\sin(2x) + \frac{1}{4}\cos(2x).\end{aligned}$$

4. Compute $\displaystyle\int \frac{1}{(\sqrt{1-x^2})^3}\,dx$.

Solution. We see a $\sqrt{1-x^2}$, and this is our signal to try a trig substitution. We set $x = \sin(\theta)$ and get $dx = \cos(\theta)\,d\theta$. Plugging everything in and integrating gives

$$\begin{aligned}
\int \frac{1}{(\sqrt{1-x^2})^3}\,dx &= \int \frac{1}{(\sqrt{1-\sin^2(\theta)})^3} \cdot \cos(\theta)\,d\theta \\
&= \int \frac{\cos(\theta)}{(\sqrt{\cos^2(\theta)})^3}\,d\theta = \int \frac{\cos(\theta)}{\cos^3(\theta)}\,d\theta \\
&= \int \frac{1}{\cos^2(\theta)}\,d\theta = \int \sec^2(\theta)\,d\theta = \tan(\theta) + C
\end{aligned}$$

And then we need to rewrite this in terms of x. From our substitution, $\sin(\theta) = x/1$, so we set up a right triangle with angle θ, where the side opposite θ is x and hypotenuse 1. We're looking for $\tan(\theta)$, so wee also need to know the adjacent side. The Pythagorean theorem gives $\sqrt{1-x^2}$, so $\tan(\theta)$ being opposite over adjacent gives $x/\sqrt{1-x^2}$. That is,

$$\int \frac{1}{(\sqrt{1-x^2})^3}\,dx = \tan(\theta) + C = \frac{x}{\sqrt{1-x^2}} + C.$$